

第 11 回 複素フーリエ級数

今回の内容

複素フーリエ級数

第 3 章 フーリエ解析 §1

定義：複素フーリエ級数（周期 $2l$ ）

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/l}, \quad c_n = \frac{1}{2l} \int_{-l}^l f(x) e^{-in\pi x/l} dx$$

$e^{i\theta} = \cos \theta + i \sin \theta$ （オイラーの公式）より、 $e^{in\pi x/l}$ は $\cos / \sin$ の代わりとして機能する

例題 6 (p. 88)

問題

次の周期 4 の関数 $f(x)$ の複素フーリエ級数を求めよ

$$f(x) = x + 1 (-2 \leq x < 2), f(x + 4) = f(x)$$

例題6 解答 (1/2)

$$l=2 \text{ より } c_n = \frac{1}{4} \int_{-2}^2 (x+1)e^{-in\pi x/2} dx$$

$n=0$ のとき：

$$c_0 = \frac{1}{4} \int_{-2}^2 (x+1) dx = \frac{1}{4} \left[\frac{x^2}{2} + x \right]_{-2}^2 = \frac{1}{4} [(2+2) - (2-2)] = 1$$

$n \neq 0$ のとき (被積分関数を分割)：

$$\int_{-2}^2 (x+1)e^{-in\pi x/2} dx = \underbrace{\int_{-2}^2 x e^{-in\pi x/2} dx}_{I_1} + \underbrace{\int_{-2}^2 e^{-in\pi x/2} dx}_{I_2}$$

例題6 解答 (2/2)

$$I_2 = \left[\frac{-2}{in\pi} e^{-in\pi x/2} \right]_{-2}^2 = \frac{-2}{in\pi} (e^{-in\pi} - e^{in\pi}) = \frac{-2}{in\pi} \cdot (-2i \sin n\pi) = 0$$

I_1 : 部分積分 $u = x$, $dv = e^{-in\pi x/2} dx$ より

$$\begin{aligned} &= \left[\frac{-2x}{in\pi} e^{-in\pi x/2} \right]_{-2}^2 + \underbrace{\frac{2}{in\pi} I_2}_{=0} \\ &= \frac{-4(-1)^n}{in\pi} - \frac{4(-1)^n}{in\pi} = \frac{-8(-1)^n}{in\pi} \quad e^{\pm in\pi} = (-1)^n \text{ を代入} \end{aligned}$$

$$C_n = \frac{1}{4} \cdot \frac{-8(-1)^n}{in\pi} = \frac{2(-1)^{n+1}}{in\pi}$$

答: $f(x) = 1 + \sum_{n \neq 0} \frac{2(-1)^{n+1}}{in\pi} e^{in\pi x/2}$

自分で解いてみよう

問6 次の周期 2 の関数 $f(x), g(x)$ の複素フーリエ級数を求めよ

$$(1) f(x) = \begin{cases} 0 & (-1 \leq x < 0) \\ 1 & (0 \leq x < 1) \end{cases}, f(x+2) = f(x)$$

$$(2) g(x) = \begin{cases} 0 & (-1 \leq x < 0) \\ x & (0 \leq x < 1) \end{cases}, g(x+2) = g(x)$$

問7 次の周期 4 の関数 $f(x), g(x)$ の複素フーリエ級数を求めよ

$$(1) f(x) = \begin{cases} 2 & (-2 \leq x < 0) \\ 0 & (0 \leq x < 2) \end{cases}, f(x+4) = f(x)$$

$$(2) g(x) = \begin{cases} 0 & (-2 \leq x < 0) \\ 2-x & (0 \leq x < 2) \end{cases}, g(x+4) = g(x)$$

問6(1) 解答

$$l = 1 : c_n = \frac{1}{2} \int_0^1 e^{-in\pi x} dx$$

$$n = 0 : c_0 = \frac{1}{2} \int_0^1 dx = \frac{1}{2}$$

$$n \neq 0 : c_n = \frac{1}{2} \left[\frac{-e^{-in\pi x}}{in\pi} \right]_0^1 = \frac{-1}{2in\pi} (e^{-in\pi} - 1) = \frac{1 - e^{-in\pi}}{2in\pi}$$

$$e^{-in\pi} = \cos(n\pi) = (-1)^n \text{ を代入 : } c_n = \frac{1 - (-1)^n}{2in\pi}$$

$$\text{答 : (1) } f(x) = \frac{1}{2} + \sum_{n \neq 0} \frac{1 - (-1)^n}{2in\pi} \cdot e^{in\pi x}$$

問6(2) 解答 (1/2)

$$l = 1 : c_n = \frac{1}{2} \int_0^1 x e^{-in\pi x} dx$$

$$n = 0 : c_0 = \frac{1}{2} \int_0^1 x dx = \frac{1}{4}$$

$$n \neq 0 : \text{部分積分 } (u = x, dv = e^{-in\pi x} dx)$$

$$= \frac{1}{2} \left\{ \left[\frac{-xe^{-in\pi x}}{in\pi} \right]_0^1 + \frac{1}{in\pi} \int_0^1 e^{-in\pi x} dx \right\}$$

$$= \frac{1}{2} \left\{ \underbrace{\frac{-e^{-in\pi}}{in\pi}}_A + \frac{1}{in\pi} \underbrace{\left[\frac{-e^{-in\pi x}}{in\pi} \right]_0^1}_B \right\}$$

問6(2) 解答 (2/2)

$$A = \frac{-(-1)^n}{in\pi} = \frac{i(-1)^n}{n\pi} \quad -1/(in) = i/n \text{ を利用}$$

$$B = \frac{-1}{in\pi}(e^{-in\pi} - 1) = \frac{1 - (-1)^n}{in\pi} \text{ なので}$$

$$(in\pi)^2 = -n^2\pi^2 \text{ より}$$

$$\frac{1}{in\pi} \cdot B = \frac{1 - (-1)^n}{(in\pi)^2} = \frac{1 - (-1)^n}{-n^2\pi^2} = \frac{(-1)^n - 1}{n^2\pi^2}$$

$$c_n = \frac{1}{2} \left(\frac{i(-1)^n}{n\pi} + \frac{(-1)^n - 1}{n^2\pi^2} \right) = \frac{i(-1)^n}{2n\pi} + \frac{(-1)^n - 1}{2n^2\pi^2}$$

$$\text{答: (2) } g(x) = \frac{1}{4} + \sum_{n \neq 0} \left[\frac{i(-1)^n}{2n\pi} + \frac{(-1)^n - 1}{2n^2\pi^2} \right] e^{in\pi x}$$

問7(1) 解答

$$l=2: c_n = \frac{1}{4} \int_{-2}^0 2 e^{-in\pi x/2} dx = \frac{1}{2} \int_{-2}^0 e^{-in\pi x/2} dx$$

$$n=0: c_0 = \frac{1}{2} \int_{-2}^0 dx = 1$$

$$n \neq 0: c_n = \frac{1}{2} \left[\frac{-2 e^{-in\pi x/2}}{in\pi} \right]_{-2}^0 = \frac{-1}{in\pi} (1 - e^{in\pi})$$

$$e^{in\pi} = (-1)^n \text{ を代入: } c_n = \frac{-1}{in\pi} (1 - (-1)^n) = \frac{(-1)^n - 1}{in\pi}$$

答: (1) $f(x) = 1 + \sum_{n \neq 0} \frac{(-1)^n - 1}{in\pi} \cdot e^{in\pi x/2}$

問7(2) 解答 (1/2)

$$l = 2 : c_n = \frac{1}{4} \int_0^2 (2-x) e^{-in\pi x/2} dx$$

$$n = 0 : c_0 = \frac{1}{4} \left[2x - \frac{x^2}{2} \right]_0^2 = \frac{1}{2}$$

$$n \neq 0 : \text{部分積分 } (u = 2 - x, dv = e^{-in\pi x/2} dx)$$

$$\int_0^2 (2-x) e^{-in\pi x/2} dx = \underbrace{\left[\frac{-2(2-x)}{in\pi} e^{-in\pi x/2} \right]_0^2}_A - \underbrace{\int_0^2 \frac{2}{in\pi} e^{-in\pi x/2} dx}_B$$

問7(2) 解答 (2/2)

$$A = 0 - \frac{-2 \cdot 2}{in\pi} \cdot 1 = \frac{4}{in\pi}$$

$$B = \frac{2}{in\pi} \left[\frac{-2e^{-in\pi x/2}}{in\pi} \right]_0^2 = \frac{-4}{(in\pi)^2} ((-1)^n - 1) = \frac{4((-1)^n - 1)}{n^2\pi^2}$$

$$\int_0^2 (2-x)e^{-in\pi x/2} dx = \frac{4}{in\pi} - \frac{4((-1)^n - 1)}{n^2\pi^2} = \frac{4}{in\pi} + \frac{4(1 - (-1)^n)}{n^2\pi^2}$$

$$c_n = \frac{1}{4} \left(\frac{4}{in\pi} + \frac{4(1 - (-1)^n)}{n^2\pi^2} \right) = \frac{1}{in\pi} + \frac{1 - (-1)^n}{n^2\pi^2}$$

答 : (2) $g(x) = \frac{1}{2} + \sum_{n \neq 0} \left[\frac{1}{in\pi} + \frac{1 - (-1)^n}{n^2\pi^2} \right] e^{in\pi x/2}$